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RESEARCH ARTICLE

# Fall Detection System on Smart Walker Based on Multisensor Data Fusion and SPRT Method

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**ABSTRACT** Traditional walkers are commonly used for the elderly in social life, which solves the basic problem of walking, but it is difficult to ensure safety when a fall occurs, and the human-computer interaction is poor. The image recognition method or the IMU sensor method fixed on the user, such as a wearable watch, is used by most of the current fall detection methods. Wearable sensors require the user's wearing operation, which is a little troublesome, and the detection accuracy is related to the way of wearing. The image recognition method requires a high-priced camera and a fixed installation position, which is unable to adapt to outdoor activities. We investigated a low-cost method of mounting the sensor on the body of the walker. We propose in this paper an improved fall detection method, namely Precondition and Limit Threshold SPRT (PLT-SPRT), and a novel fall detection system on the smart walker based on PLT-SPRT. The signals of the upper and lower limb sensors are fused based on the Kalman filter algorithm, and the admittance control parameters are obtained through the system identification method. In this study, the improved sequential probability ratio test algorithm is used to set the null hypothesis and the alternative hypothesis, construct the likelihood ratio and optimize the decision function, which is used to judge whether falls occur. The system is simulated by Matlab software, the user intention after fusion is more accurate, and the optimized decision function is judged accurately. Verified by the embedded system based on STM32 of the smart walker equipment in the real world, it can accurately identify the fallen state, with low detection delay, and the fallen state is detected about 160ms earlier than the traditional threshold-based detection algorithm, at the same time, the accuracy is higher than 94.9%, which meets the high real-time requirements of fall detection and is the ideal solution for smart walkers.

**INDEX TERMS** Admittance control, data fusion, fall detection, smart walker, SPRT.

## I. INTRODUCTION

As the elderly grow older, the functions of various organs and physical functions decline year by year. Osteoporosis, muscle atrophy, and other problems are not conducive to the normal walking of the elderly [1]. At present, there are two types of smart devices to solve the walking problem of the elderly and the disabled: electric wheelchairs and smart walkers. Electric wheelchairs can completely replace human legs to complete basic walking [2], while smart walkers can be used to assist with walking, and carry out related rehabilitation training, which is a key research direction in the field of medical devices [3]. Falling is a significant challenge for the elderly group that can reduce life expectancy. Approximately 35% of people (over 65 years old) have one or more falls each year, which can lead to fatal injuries [4], [5]. This percentage includes 87% of users with walking support systems, who are treated at hospital emergency departments for injuries from falls that involve walkers and canes [5]. Elderly people have a risk of falling due to many factors including decreased

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physical and psychological conditions, loss of consciousness, vision problems, joint problems, respiratory problems, heart failure, and other health problems. Especially falls may cause enormous medical expenses and heavy social burdens. Most the deaths from falls are caused by failure to get timely and effective medical assistance. When a fall happens, most elderly fallers cannot rise on their own or send out a warning signal, and the length of time from falling to receiving effective medical assistance is crucial to the rehabilitation and health of the elderly. Thus, a fruitful of fall detection methods have been proposed to deal with this diffculty.

Smart walkers detect the surrounding environment and perform human-computer interaction by installing various sensors on them, such as the research of Ragaj *et al.* [6] mentioned that a good control algorithm can help users use the walker better. Unlike electric wheelchairs, smart walkers do not completely hand over the control of the lower limbs to rehabilitation equipment, but the users themselves control the rehabilitation equipment to complete their walking [7]. Falls may occur during this process, so safety must be ensured. Whether it is an elderly or a disabled person, the consequences of an accidental fall may be serious. Therefore, designing an effective and reliable fall detection algorithm for smart walkers is necessary. When a fall is detected, the walker should be braked urgently to help the user return to a normal walking posture.

Common fall detection methods include image-based detection and wearable sensors [8]. Image-based detection methods mainly use neural networks and other algorithms to classify and recognize human walking postures and falling postures collected by cameras, such as the fall detection framework proposed by Chaudhari *et al.* [9]. But cameras are easily affected by the surrounding environment, such as lighting conditions. Taghvaei *et al.* [10] use the camera to predict falls instead of directly detecting falls, and later Taghvaei *et al.* [11] provided an image-based fall detection research, using positive state distribution to judge the user's walking state and hidden Markov models (HMMs) for fall detection. The methods of detecting falls using wearable sensors require the installation of corresponding sensors on the user. Young-Hoon Nho *et al.* [12] propose an automatic fall detection method for a wearable device. They propose a novel generative adversarial network (GAN-) based fall detection method using a heart rate sensor and an accelerometer. They introduce a compelling GAN-based anomaly detection partially surrounded by User Initial information features (UI-GAN). Yves M. Galvão *et al.* [13] provide a multimodal approach using deep learning for fall detection by proposing different topologies of a multimodal convolution neural network, which is trained to detect falls based on RGB images and information from accelerometers, making it much more scalable. Palmerini *et al.* [14] installed accelerometer sensors to detect the movement of human joints, thereby calculating the walking posture of the human body. Some joints may not be able to install sensors, making fall detection difficult to perfect. There are also research on installing various inertial sensors in other parts [15]–[17]. Many methods finally achieve satisfactory goals [18], but not all of them are suitable for outdoor movable walkers, such as the research by Skubic *et al.* [19], who proposed several methods for fall detection by using the Microsoft Xbox Kinect, and this system can only be used in fixed scenarios. In their research, they installed a depth camera system directly in the apartments of 94 elderly people.

Sen Qiu *et al.* [20] provide the latest research progress on Multi-sensor information fusion based on machine learning for real applications in human activity recognition. This review introduces common wearable sensors, smart wearable devices, and key application areas. This review attempts to provide a more comprehensive survey of the most important aspects of multi-sensor applications for human activity recognition, including those recently added to the field for unsupervised learning and transfer learning. Moreover, the open research issues that need further research and improvement are identified and discussed. Xie *et al.* [21] directly used smart watches to detect falls of the elderly. Users only need to wear a watch integrated with the corresponding sensor. However, their detection accuracy is not very high, and if the elderly do not wear a watch, for example, it is not very convenient to use a walker to go to the toilet after getting up in the morning. Herti *et al.* [22] used LiDAR fixed indoors to detect falls, which is not suitable for outdoor travel, and multi-threaded/3D LiDAR is more expensive than cameras and other sensors.

Floor-based sensors such as Ground Reaction Force (GRF) sensors and pressure sensors observe the force extracted by human feet to observe the fall. Serra *et al.* used sensors to detect falls based on-ground response, but this method cannot be applied to outdoor mobility walkers [23]. Various sensors are deployed ranging from ultrasonic, motion sensors, sound sensors, radars, and others to detect fall incidents in a room [24], [25].

Among the existing fall detection methods, Nyan *et al.* [26] predict falls by comparing the angle and angular velocity of the thigh in various fall behaviors and daily normal behaviors, Shi *et al.* [27] and Shan and Yuan [28] predict falls by extracting features that collected by accelerometers and gyroscopes through using support vector machines. Wu and Xue [29] and Bourke *et al.* [30] detect falls by setting the vertical velocity threshold of humans. They detect falls by fixing sensors on the user's body, and the algorithms are complex, these processes need a lot of training and calculation.

This study aims to design a multi-sensor fall detection method that can be used in various light conditions without requiring special installation and wear by the user, and the cost is relatively low. Relevant sensors are installed on the body of the smart walker to sense the user and the situation around him. The user does not need to wear the devices or sensors and is not affected by light. It can be used outdoors at a low cost.

In this paper, we propose a fall detection method suitable for smart walkers, which uses force sensors on the handle

of the walker to detect the user's upper limb force signals, as well as ranging sensors (ultrasonic and Laser Range Scanners (LRS)) to detect the user's lower limb movement signal, and the multi-sensor data fusion is performed by the Kalman filter algorithm to obtain the user's intention velocity.

Finally, an improved Sequential Probability Ratio Test (SPRT) method is used for the fused data to detect whether a fall occurs. To measure the strength of the upper limb in real-time and adapt to the various angles of the user holding the handle, we also creatively build a new handle.

The contributions of this paper can be summarized as follows:

(1) We propose a new method and device for upper limb strength detection.

(2) By analyzing the motion of the upper limbs, we establish a system identification method based on the two-parameter model to identify the admittance control model.

(3) We propose a data fusion method based on the Kalman filter algorithm to fuse the sampling signals of upper and lower limb sensors.

(4) In addition, we propose a fall detection algorithm named the PLT-SPRT method, which optimizes its decision function and increases its practicality in practical applications, so that it can identify falls faster and improve the recognition speed.

As a medical rehabilitation device, the safety and status detection of the walker is very important. The external sensors used in this study are also used for other functions, such as admittance to control the operation of the smart walker, detection of whether the user is on an uneven road, or if there is an obstacle ahead, etc.

The new handle can identify the strength of a walker user's upper limb in real-time. This algorithm can quickly detect the user's falling behavior with high accuracy. The smart walker developed in this research has a semi-enclosed structure and has safety functions such as active braking. Therefore, although the user's fall does not cause immediate danger, this study still achieves an average fall detection time of 244ms, which is about 160ms earlier than the traditional SPRT method to detect fall states. The experimental results show that the method has fast detection velocity and high accuracy, which provides a guarantee for the safety of smart walkers.

The rest of this paper is organized as follows: Section III provides a new handle to analyze the user's upper limbs first, combined with lower limb motion analysis, a multi-sensor fusion algorithm is presented, and then elaborates on fall detection algorithm in fall detection. The experiments are presented in Section IV. Finally, we present our concluding remarks and share the directions of our ongoing work in Section V.

## II. RELATED WORKS

Various advanced fall detection techniques and related research are developing rapidly. The focus of this technology is to detect falls quickly and accurately and to minimize the consequences of falls. Currently, falls are generally detected by sensors and cameras [8]. The sensor detection method can be distinguished as wearable and non-wearable, and to make it easier for users to get started, this study uses a combination of the advantages of wearable and non-wearable devices to fix the sensors on the smart walker. This way not only simplifies the use of the user but the smart walker can be moved at any time.

Based on previous research, Zhao *et al.* [31] built a smart walker with a hybrid traditional method and roller design to detect falls by detecting lower limb gait. However, they did not consider the variation of upper limbs.

In this study, haptic handles for upper limbs are added, referring to the work of Ram' on *et al.* [1], [32], their work presents a pair of haptic handles that incorporates both vibration haptic feedback and a novel skin-stretch-and-slip haptic feedback. This study developed a handle that integrates the force sensors matrix.

Admittance control is widely used in the field of robot control. This study refers to the Adaptive Hybrid System Framework based on a feed-forward neural network [33], and the navigation strategy which uses an admittance controller to generate haptic and visual signals in the smart walker proposed by Jiménez *et al.* [34].

There are many researches related to multi-sensor signal fusion in the field of smart walker. Feng *et al.*'s work on Kalman filtering for multi-sensor processing [35] shows the way the data is processed with least squares (LS).

The study by Li *et al.* [36] on the remaining useful life of the multi-sensor data fusion model also helped to make some optimizations for the multi-sensor data fusion in this study. This study refers to the research [29], [37], [38] on Kalman filtering, to remove noise and unwanted signals from time-domain signal data.

Wald proposed sequential probability ratio test (SPRT) in 1947 [39]. The statistical decision is made in SPRT by using the sequence observations in each signal channel and the likelihood ratio test in each cycle step. SPRT technology is under development, Wu *et al.* [40] proposed Soft truncation (ST)-SPRT, which can improve detection performance in terms of detection performance. This study refers to some ideas of soft-truncated SPRT.

## III. METHODOLOGY

The design diagram of the smart walker is as follows:

As shown in Fig. 1(a) and (b), each handle of the upper limb is composed of a matrix of 8 thin-film pressure sensors, which are used to collect the force signal of the user's hand. Ranging sensors are installed on the user's waist, knees, and the upper part of ankles. The smart walker is designed to be a four-wheeled chassis which is currently used in the market. The main structures include front-wheel steering and rear-wheel differential drive, two-wheel drive plus two universal wheels, and four-wheel drive as shown in Fig. 1(c). Currently, ultrasonic ranging sensors are used, and we also have

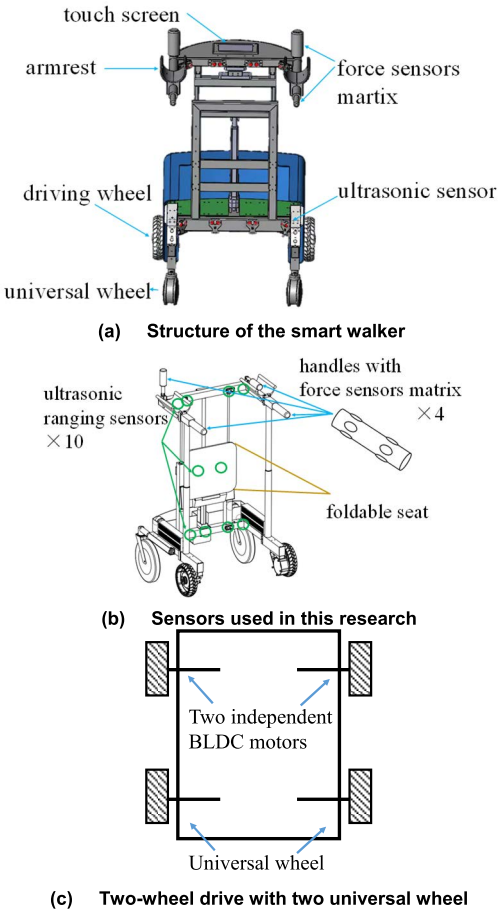



**FIGURE 1.** Structure of the smart walker and sensors used in this research.

installation locations for laser ranging sensors. 10 ranging sensors with force sensors matrix in various positions are mounted on smart walker as Fig. 1(b). There are 4 handles on the smart walker, providing two ways of use, prone mode and push mode.

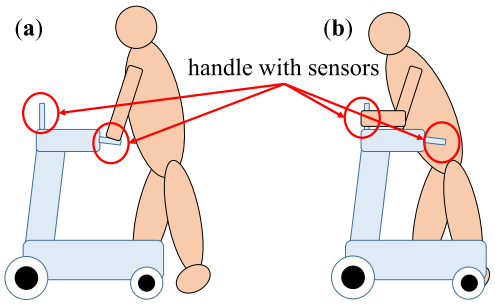



**FIGURE 2.** Two ways to use smart walker. (a) Push mode. (b) Prone mode.

The prone mode means that the user lies on the body of the walker and uses the front two armrests; the push mode is similar to pushing a cart, using the rear two armrests, with force sensors installed on all four armrests to recognize the user's intention.



**FIGURE 3.** Schematic diagram of smart walker folding.

A soft cushion is provided on top of the walker's body when the user is lying down, and a folding seat and footrest are designed so that the walker can be changed into a wheelchair form for rest when the user is tired. The overall height of the walker is adjustable.

The above content introduces the overall structure and sensor distribution of the smart walker, and the next Section A will introduce the upper limb motion analysis, Section B will introduce system identification of upper limb sensors, Section C will introduce lower limb analysis, and Section D will introduce multi-sensor data fusion, Section E and Section F will introduce SPRT and PLT-SPRT.

### A. UPPER LIMB MOTION ANALYSIS

The difference between smart walkers and traditional electric wheelchairs is that the wheelchairs control the movement of the wheelchair through a ''remote controller''. There is no human-computer interaction between the hands and the wheelchairs. The user presses the ''go forward'' button and the controller executes after the command is reached while the smart walkers install force sensors on the armrest to sense the user's force, to identify the user's intention through the relevant algorithm. If the user wants to accelerate, the overall force on the armrest can be increased, so that the force sensors detect the change of the user's force, and the controller drives the motor to complete the corresponding acceleration movement. The principle of deceleration is similar to acceleration, and the user just needs to reduce the force.

The way of using the smart walker increases the involvement of hands and is simple to operate. Electric wheelchairs need to press multiple buttons to solve complex situations. Pressure sensor handles built specifically for smart walkers can be presented in many ways, but if we use a single sensor, or if we use multiple sensors but their force direction is the same, then it won't work for smart walkers. The handles built with pressure sensors for smart walkers can be presented in many ways, but if we use a single sensor, or if we use multiple sensors but their force direction is the same one, then it won't work for smart walkers.

The smart walker in this work can follow the user's hands to complete related actions, and the control method is more flexible and fast in processing. However, the user's control of the armrest is not simply a matter of increasing or decreasing

the interactive force, the interactive force has strict requirements on the installation position and quantity of the sensors on the armrest. Taking the push mode as an example, the installation method of the sensors on the armrest is shown in the following Fig. 4:

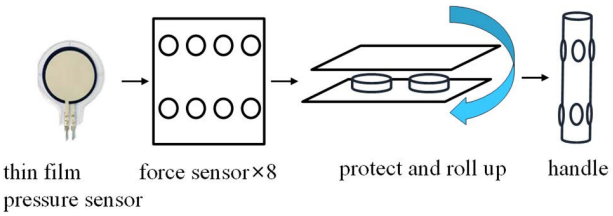



**FIGURE 4.** Self-build force sensor handle.

The force sensors are installed on the armrest of the smart walker according to the characteristics of the force of the user's palm. The interactive force is obtained according to the force of each force sensor, and the intention of the upper limb is identified, such as the movement direction, acceleration, deceleration, etc. Taking forward motion as an example, analyze the force of the armrest, the user's forward direction is the same as the positive direction of the $R_y$-axis of the walker coordinate system, and the following coordinate system of the walker is established as shown in Fig. 5:

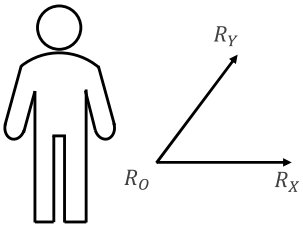



**FIGURE 5.** Walker coordinate system.

Analyze the force of the armrest:

As shown in Fig. 6(a), let the angle between the armrest or handle of the walker and the horizontal direction be $\theta$. When moving forward, the combined force of the eight force sensors in the vertical direction of the left armrest is $F_L$, the direction is perpendicular to the armrest downward, the right hand is similar to the left hand, and the resultant force is $F_R$. The relationship between them is shown in Fig. 6(b), the component forces in the positive direction of the $R_y$-axis are as formula (1):

$$\begin{cases} F_{yl} = F_L \cdot sin\,\theta \\ F_{yr} = F_R \cdot sin\,\theta \end{cases} \tag{1}$$

Calculate the force of the left hand and right hand in the $R_x$-axis and $R_y$-axis directions, the final combined forces $F_L$ and $F_R$ in the $R_x$-axis and $R_y$-axis directions can be calculated as formula (2):

$$\begin{cases} F_x = F_{xl} + F_{xr} \\ F_y = F_{yl} + F_{yr} \end{cases} \tag{2}$$

Analyze the force of the armrest:

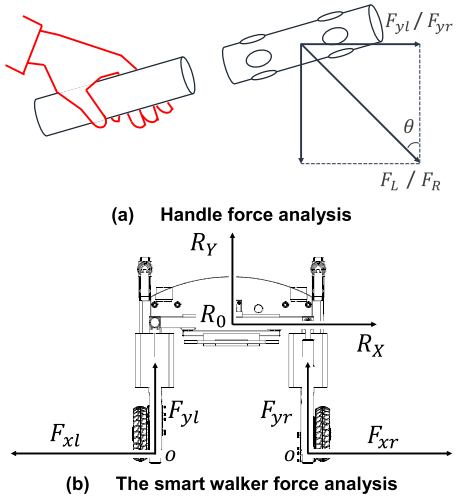



**FIGURE 6.** Schematic diagram of handle and the smart walker force analysis.

After obtaining the magnitude and intention direction of the interactive force, it is also necessary to correspond to the control of the smart walker, there must be a certain corresponding relationship between the force and the velocity. When the armrest of the walker is subjected to the interaction force from the user, the user's intention is solved according to the relevant algorithm [32], and finally the main controller inputs this velocity parameter into the motor driver to complete the human-machine interaction with the smart walker.

### B. TWO-PARAMETER MODEL-BASED SYSTEM IDENTIFICATION OF UPPER LIMB SENSORS

Admittance control is a commonly used control method in robot control. The input is a force signal and the output is a position signal. The interaction force between the smart walker and the user is measured, and the velocity of the walker can be obtained by inputting the interaction force into the system admittance control model. The dynamics model of admittance control is adopted as same as in [33] and [34] as formula (3):

$$f = M\ddot{x} + C\dot{x} + K\Delta x \tag{3}$$

$f$ is the interaction force, $M$ is the inertia of the walker, $C$ is the damping coefficient, $K$ is the stiffness coefficient, and $x$ is the displacement vector. Generally, $K$ is relatively small and can be ignored, so the model becomes formula (4):

$$f = M\ddot{x} + C\dot{x} \tag{4}$$

Since the controller needs the velocity signal instead of the displacement signal when driving the motor, according to the relationship between the displacement and the velocity, the model is further transformed into formula (5):

$$f = M\dot{v} + Cv \tag{5}$$

Taking the Laplace transformation on both sides of formula (5):

$$\frac{V(s)}{F(s)} = \frac{1}{Ms + C} = G(s) \tag{6}$$

It can be seen that the typical first-order inertial link can be obtained by simply transforming the transfer function $G_{(S)}$ of the admittance control as formula (7):

$$\frac{k}{Ts + 1} = \frac{1}{Ms + C} \tag{7}$$

Then $M$ and $C$ can be represented by $T$ and $k$ of the first-order inertial elements as formula (8):

$$\begin{cases} k = \dfrac{1}{C} \\ T = \dfrac{M}{C} \end{cases} \tag{8}$$

The time-domain in the model of admittance control is as formula (9):

$$\frac{dV(t)}{dt} + V(t) = k \cdot F(t) \tag{9}$$

Discretizing it, there is formula (10)

$$T \cdot \frac{V(n) - V(n-1)}{\Delta t} + V(n) = k \cdot F(n) \tag{10}$$

The relationship between the admittance control force and the velocity is given as formula (11):

$$V(n) = k \cdot \frac{\Delta t}{\Delta t + T} \cdot F(n) + \frac{T}{\Delta t + T} \cdot V(n-1) \tag{11}$$

It can be seen from above that $\Delta t$ is known as the sampling time interval of the system, and the relationship between the interaction force and the velocity can be calculated by the values of $T$ and $k$. Therefore, in this study, the two-parameter system identification is used to identify the values of $T$ and $k$.

The transfer function of the two-parameter model can be calculated by the formula (12):

$$G(s) = \frac{k}{T_{ar}s + 1} \tag{12}$$

where $k$ is the object gain and $T_{ar}$ is the time constant. When the input is a unit step signal, the response curve of the system is shown in Fig. 7:

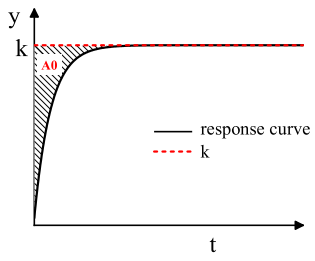



**FIGURE 7.** Two-parameter model system response curve.

This gives the value of $k$:

$$k = y(\infty) \tag{13}$$

When the input is a step signal, there are:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{k}{T_{ar}s + 1} \tag{14}$$

Then:

$$Y(s) = G(s) \cdot U(s) = \frac{k}{T_{ar}s + 1} \cdot \frac{1}{s} \tag{15}$$

Taking the Laplace transformation on both sides of the formula (16):

$$y(t) = L^{-1}(Y(s)) = k(1 - e^{-\frac{t}{T_{ar}}}) \tag{16}$$

From the response curve of the system, the area of the shaded part can be found as formula (17):

$$S = \int_0^\infty k - y(t)dt = \int_0^\infty k \cdot e^{-\frac{t}{T_{ar}}} dt = T_{ar} \cdot k \tag{17}$$

Then $T_{ar}$ can be calculated as formula(18):

$$T_{ar} = \frac{A_0}{k} \tag{18}$$

The values of $k$ and $T_{ar}$ are already known, and $k$ and $T_{ar}$ in the two-parameter model correspond to the values of $k$ and $T$ in the admittance control model as Fig.8 (a), and the system identification process is shown in Fig.8 (b).

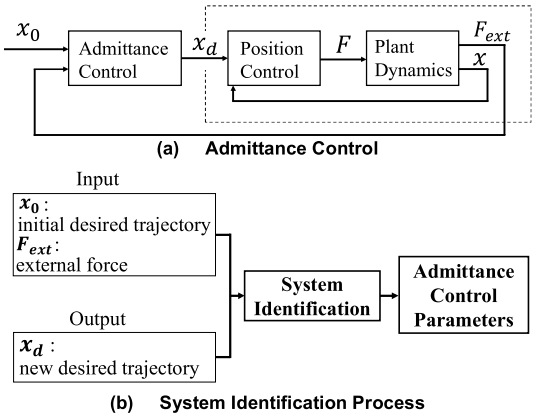



**FIGURE 8.** Diagram of solving admittance control parameters.

In Admittance Control, the plant is position-controlled [33]. The admittance controller outputs the new desired trajectory $\boldsymbol{x}_d$, which is processed by the PD controller and outputs force signal $\boldsymbol{F}$. Hence the controller is designed as shown in Fig.8 (a).

### C. LOWER LIMB MOTION ANALYSIS

Elderly people who need to use a walker to walk are different from healthy people, as shown in Fig. 9:

According to the above analysis, when elderly people walk forward with the smart walker, the midpoint of the projection of the legs on the ground is taken as the reference point, and the inertial coordinate system shown in Fig. 10 is established. The origin of the coordinates $W_o$ coincides with the reference point at the initial moment:

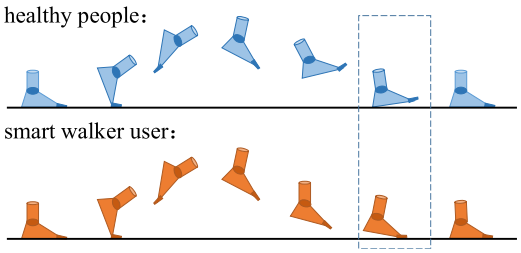



**FIGURE 9.** Differences between healthy people walking and walker users walking.

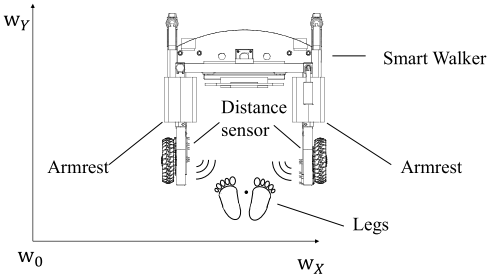
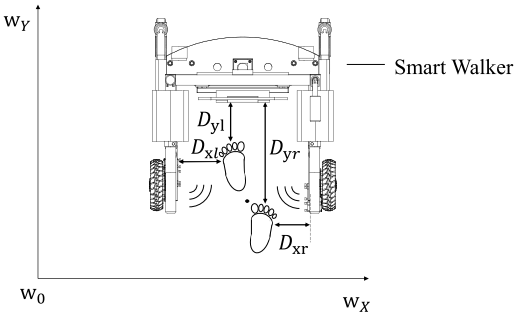
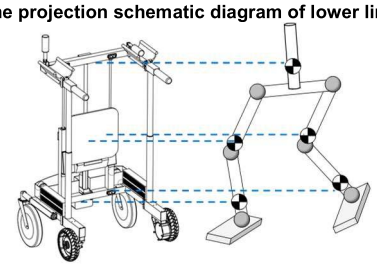



**FIGURE 10.** Inertial coordinate system.




(a) Plane projection schematic diagram of lower limb signal



(b) Stereoscopic diagram of lower limb signal

**FIGURE 12.** Schematic diagram of lower limb signal.

In the process of moving forward, the movement of the lower limbs/legs should swing alternately. The whole process can be divided into 4 steps:

1) Swing the left leg forward to the ground;
2) Swing the right leg to a position flush with the left leg;
3) Continuing to swing the right leg until it hits the ground;
4) Swing the left leg to a position flush with the right leg.

The above process is a complete walking process, and its repeated process is analyzed each time. Use the projection of the legs on the ground to draw a schematic diagram, as shown in Fig. 11:

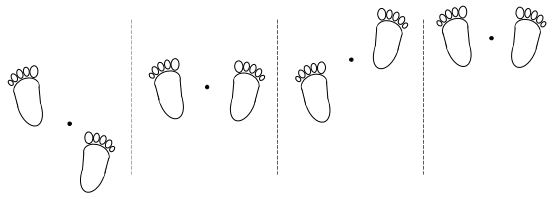


**FIGURE 11.** Walking process. (a) step 1, (b) step 2, (c) step 3, (d) step 4.

According to the coordinate system established in Fig. 10, the distance between the legs and the edge of the smart walker in each step can be measured and converted into the inertial coordinate system using the laser ranging sensors, taking step 1 as an example:

As shown in Fig. 12, $D_{xl}$, $D_{xr}$, $D_{yl}$, and $D_{yr}$ can be measured by the distance measuring laser sensors, where $D_{xl}$ and $D_{xr}$ are the distance between the left leg and the left side of the smart walker and the distance between the right leg and the right side of the smart walker, $D_{yl}$ and $D_{yr}$ are respectively the distance between the left and right legs from the front side of the smart walker. To simplify the model, only the kinematics model of forwarding walking is now analyzed. The users of smart walkers are mostly elderly and disabled people, the amplitude of each swing of the legs is very small. The distance traveled by the walker is equal to the distance traveled by the midpoint of the projection of the user's legs on the ground. In the initial state, $D_{yl}$ is equal to $D_{yr}$. When the user is in step 1, swinging his left leg forward, $D_{yl}$ decreases, and $D_{yr}$ increases. $H_O(x_h, y_h)$ is the midpoint of the projection of the legs. Let the width of the body of the intelligent walker be $W$, $W_O(0, 0)$ and $H_O(0, 0)$ coincide at the initial moments, and the initial values of $D_{xl}$, $D_{xr}$, $D_{yl}$, and $D_{yr}$ are $D_{xl0}$, $D_{xr0}$, $D_{yl0}$ and $D_{yr0}$, there is formula (19):

$$\begin{cases} x_h = \dfrac{W - D_{xl} - D_{xr}}{2} + D_{xl} - \left(\dfrac{W - D_{xl0} - D_{xr0}}{2} + D_{xl0}\right) \\ y_h = D_{yl} - D_{yl0} \end{cases} \tag{19}$$

The formula (19) is the coordinate analysis of step 1. The coordinate algorithm in the $W_X$-axis direction of the subsequent steps is the same as that in step 1, and the coordinate algorithm method in the $W_Y$-axis direction is a little different. Because the user is moving forward along the $W_Y$-axis, the ordinate should gradually increase, so the accumulation should be performed according to the ordinate value at the previous moment.

### D. MULTI-SENSOR DATA FUSION

After the user's motion coordinates are obtained, the displacement and velocity can be obtained. The displacement is measured by the lower limb ranging laser sensors, and the velocity is calculated by the upper limb force sensors through

the admittance control model. Due to the uncertainty of the measurement by a single kind of sensor, the high reliability of the data cannot be guaranteed, so multi-sensor data fusion is performed [36]. The data collected by different types of sensors are fused and processed to obtain higher reliability data. Only fall detection in the forward direction is included in this study.

The data types that the smart walker needs to fuse are displacement and velocity, which fit the typical application range of the Kalman filter [35]. Multi-sensor data fusion is performed using Kalman filtering, the historical data measured by sensors is saved, and then the least-squares method is used for linear fitting [43], and the fitted function is used as the subsequent prediction function. The predicted value of the Kalman filter is provided by the value of the fitted function, and let the fitted function be $y = a+bx$, the measurements at $n$ observation points $(x_1,x_2,\ldots,x_n)$ are $(y_1, y_2,\ldots,y_n)$, the error is $d_i = y_i - (a + bx_i)$, $i = 1, 2, \ldots, n$. Let the total error be $D_i$, then we can get formula (20):

$$D_i = \sum_{i=1}^{n} d_i^2 \tag{20}$$

The fitting effect is the best when the value of $D_i$ is the smallest, and the partial derivative of $D_i$ can be obtained by formula (21):

$$\begin{cases} \dfrac{\partial D_i}{\partial a} = -2(\sum\limits_{i=1}^{n} y_i - na - b\sum\limits_{i=1}^{n} x_i) \\ \dfrac{\partial D_i}{\partial b} = -2(\sum\limits_{i=1}^{n} x_i y_i - a\sum\limits_{i=1}^{n} x_i - b\sum\limits_{i=1}^{n} x_i^2) \end{cases} \tag{21}$$

Let $\partial D_i/\partial a$ and $\partial D_i/\partial b$ be equal to 0, according to formula (22):

$$\begin{cases} \bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n} \\ \bar{y} = \dfrac{\sum_{i=1}^{n} y_i}{n} \end{cases} \tag{22}$$

$a$ and $b$ can be calculated as formula (23):

$$\begin{cases} a = \bar{y} + b\bar{x} \\ b = \dfrac{\sum_{i=1}^{n} x_i y_i - \bar{y}\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} x_i^2 - \bar{x}\sum_{i=1}^{n} x_i} \end{cases} \tag{23}$$

So far, the linear fitting function $y = a + bx$ has been solved, and the $n+1$ predicted value can be obtained according to the first n measured values, and then the Kalman filter algorithm is used to perform the optimal estimation at the time $n + 1$.

The state equation and measurement equation of the system is utilized as formula (24) [37]:

$$\begin{cases} \boldsymbol{X_k} = \boldsymbol{A X_{k-1}} + \boldsymbol{w_{k-1}} \\ \boldsymbol{Z_k} = \boldsymbol{H X_k} + \boldsymbol{v_k} \end{cases} \tag{24}$$

And:

$$\boldsymbol{X_k} = \begin{bmatrix} x_t & y_t & v_{xt} & v_{yt} \end{bmatrix}^T \tag{25}$$

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{26}$$

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{27}$$

where $x_t$ and $y_t$ correspond to the motion displacement in the $W_X$-axis and $W_Y$-axis directions calculated by the ranging sensors, respectively, $v_{xt}$ and $v_{yt}$ correspond to the response velocity of the upper limb force in the $W_X$-axis and $W_Y$-axis directions of the admittance control model, $A$ is the state transition matrix, $\omega$ is the process noise and $v$ is the measurement noise.

These multi-sensor data are fused based on the Kalman filter algorithm:

Predicted state vector as formula (28):

$$\hat{\boldsymbol{X}}_{\boldsymbol{k}}^{'} = \boldsymbol{A}\hat{\boldsymbol{X}}_{\boldsymbol{k-1}} \tag{28}$$

Prediction error covariance as formula (29):

$$\boldsymbol{P}_{\boldsymbol{k}}^{'} = \boldsymbol{A P_{k-1} A^T} + \boldsymbol{Q} \tag{29}$$

Update Kalman gain as formula (30):

$$\boldsymbol{K_k} = \boldsymbol{P_k^{'} H^T} (\boldsymbol{H P_k^{'} H^T} + \boldsymbol{R})^{-1} \tag{30}$$

Update state vector as formula (31):

$$\hat{\boldsymbol{X}}_{\boldsymbol{k}} = \hat{\boldsymbol{X}}_{\boldsymbol{k}}^{'} + \boldsymbol{K_k}(\boldsymbol{Z_k} - \boldsymbol{H}\hat{\boldsymbol{X}}_{\boldsymbol{k}}^{'}) \tag{31}$$

Update error covariance as formula (32):

$$\boldsymbol{P_k} = (\boldsymbol{I} - \boldsymbol{K_k H})\boldsymbol{P_k^{'}} \tag{32}$$

So far, more accurate displacement and velocity have been obtained after multi-sensor data fusion.

### *E. SEQUENTIAL PROBABILITY RATIO TEST*

After the user's motion coordinates are obtained, the displacement and velocity can be obtained.

The Sequential Probability Ratio Test (SPRT) is an important branch of mathematical statistics. Instead of specifying the sample size in advance, the method draws a small number of samples first and then performs a series of tests based on these samples before deciding whether to continue sampling, solving the sample size problem in the probability ratio test, and improving the speed of the test.

The sequential probability ratio test is also a hypothesis test [39], [45]. Select the appropriate null hypothesis $H_0 : \theta = \theta_0$ and alternative hypothesis $H_1 : \theta = \theta_1$, when the null hypothesis is true, the judgment result rejects the null hypothesis and chooses the alternative hypothesis. This is the first error called misreporting, and the probability of occurrence is $\alpha$; On the contrary, when the null hypothesis is false, the judgment result accepts the null hypothesis and rejects the

alternative hypothesis. This is the second error called under-reporting, and the probability of occurrence is $\beta$. The random variable $x_i$ obeys the positive state distribution $N\left(\mu, \sigma^2\right)$, the probability density functions of the null hypothesis and the alternative hypothesis can be expressed as formula (33) and formula (34):

$$p_{0i} = \frac{1}{\sigma\sqrt{2\pi}} exp(-\frac{1}{2\sigma^2}(x_i - \mu_0)^2) \tag{33}$$

$$p_{1i} = \frac{1}{\sigma\sqrt{2\pi}} exp(-\frac{1}{2\sigma^2}(x_i - \mu_1)^2) \tag{34}$$

Among them, $i = 1, 2, \ldots, n$ is the sampling ordinal number of the data, and $\mu$ corresponds to $\theta$ of the null hypothesis and the alternative hypothesis, the likelihood ratio can be expressed as formula (35):

$$\lambda_i = \frac{\prod_{i=1}^{n} p_{1i}}{\prod_{i=1}^{n} p_{0i}} \tag{35}$$

For the convenience of calculation, the logarithm operation is usually taken on $\lambda_i$, and the decision formula is formula (36):

$$\Delta = ln\lambda_i = ln\frac{\prod_{i=1}^{n} p_{1i}}{\prod_{i=1}^{n} p_{0i}} = \sum\nolimits_{i=1}^{n} ln\frac{p_{1i}}{p_{0i}} \tag{36}$$

Thresholds $A$ and $B$ are related to $\alpha$ and $\beta$ as formula (37):

$$\begin{cases} A = \frac{1-\beta}{\alpha} \\ B = \frac{\beta}{1-\alpha} \end{cases} \tag{37}$$

Correspondingly, after taking the logarithm, the thresholds become $lnA$ and $lnB$, and the judgment relationship of the sequential probability ratio test is as formula(38):

$$\begin{cases} \Delta < ln\, B \\ ln\, A > \Delta > ln\, B \\ \Delta > ln\, A \end{cases} \tag{38}$$

When $\Delta < ln\ B$, stop the test and receive the original result; when $ln\ A > \Delta > ln\ B$, it's unable to determine, continue sampling; and when $\Delta > ln\ A$, stop the test and accept the alternative hypothesis.

### F. FALL DETECTION ALGORITHM FOR SMART WALKERS BASED ON IMPROVED SPRT

The most important thing in using smart walkers is to ensure the safety of the elderly in the walking process. The elderly and the disabled people have poor lower limb movement ability, and are very prone to falls, resulting in irreversible consequences. Therefore, smart walkers must have a fall detection function. The usual occurrence of falls can be divided into two kinds: forward falls and side-to-side falls. This study investigates an improved SPRT-based fall detection algorithm. The detection principles of these two fall situations are roughly the same, and the forward fall situation is now analyzed.

When the user falls forward, the upper limbs show that the body leans forward rapidly, and the interaction force of the hands-on the armrest increases; the lower limbs show that the distance between the legs and the body of the walker increases, and the swing speed increases. According to the above analysis, a large number of experiments are conducted to collect upper and lower extremity data under normal and falling conditions, and after filtering, they are used as the null hypothesis and the alternative hypothesis. The upper limb and lower limb data are fused with the multi-sensor data described above, and the fused data is subjected to a sequential probability ratio test, the likelihood ratio is constructed, and a decision function is established to determine whether a fall occurs.

ESPRT[50] is a reasonable method to complete fall detection. Using both entropy and SPRT, the method can reduce the false positive rate, but the speed of the detection may not keep up with the design requirements of fall detection.

The initial value of the decision function of the traditional SPRT method can be zero. When the value of the decision function is greater than $lnA$ or less than $lnB$, the termination condition is satisfied, and the test is stopped immediately. The decision function value can be re-judged from zero according to the specific selection of the null hypothesis and the alternative hypothesis. For example, in the fall detection, the null hypothesis is normal walking, and the alternative hypothesis is a fall. When the value of the decision function is greater than the threshold $lnA$, the test is stopped and a fall judgment is given; when the value of the decision function is less than the threshold $lnB$, the test is stopped and a judgment of the normal state is given. However, for the fall detection of smart walkers, the most important thing is to detect falls quickly and accurately, so it is not necessary to give the judgment of the normal state frequently, and the frequent reset of the decision function value may cause misreporting. Therefore, the determination of less than the threshold can be optimized. Since the decision function of the SPRT method will accumulate negative values in the normal state when calculating the likelihood ratio, it will accumulate positive values for some time in the process of detecting a fall, resulting in delay in detection, so it can be optimized to eliminate the detection delay.

$$\Delta'(k) = \begin{cases} \Delta(k) & \Delta(k) \geq ln\ B \\ ln\ B & \Delta(k) < ln\ B \end{cases} \tag{39}$$

where $k$ is the number of the sampling sequence in formula(39).

During the use of the smart walker, if the user suddenly falls, sometimes the lower limbs do not drop rapidly, and the distance does not change suddenly. For example, when the user falls, the body will get down slowly. Because the human body is a flexible associative body, it has a basic anti-mutation ability. Because the upper limbs are more flexible, they can perform the stress action of falling faster than the lower limbs. To detect falls in the shortest time and take emergency braking and other protective measures, the algorithm can give higher weight to upper limbs when judging falls.

To realize the above design, SPRT detection needs to be modified to add preconditions and weights, which can be

expressed as formula (40) and formula (41):

$$\Delta^{''}(k) = \Delta'(k)C + ln\,A \cdot (1 - C) \tag{40}$$

$$\Delta^{''}(k) = \Delta'(k) \cdot \prod_{i=1}^{n} (1 - r_i) + ln\,A \cdot \sum_{i=1}^{n} r_i \tag{41}$$

$C$ is the state of judging the occurrence of the positive precondition, and if there are more than 2 possible options, $r_i$ is the state of judging the occurrence of the $i$ precondition:

$$\begin{cases} r = 1; & \textit{When the preset condition is triggered}; \\ r = 0; & \textit{normal situation}; \end{cases} \tag{42}$$

According to formula(42), when the $i$ precondition is triggered, $\Delta''(k) = lnA$, which immediately triggers the critical value for SPRT fall detection. If the precondition is triggered during using the smart walker, the data signal is fused, and after the improved SPRT algorithm, it is detected that the falling threshold has been reached. Since fall detection requires a fast response, it is more reasonable to trigger it directly instead of waiting for algorithm iterations.

In this algorithm, only the condition that the two handles of the upper limb of the walker are abnormal during normal travel is detected, at that time, the handles are completely released or there is a significant rise in the instantaneous force. Here $n = 2$,

$$\Delta^{''}(k) = \Delta'(k) \cdot (1 - r_1) \cdot (1 - r_2) + ln\,A \cdot (r_1 + r_2) \tag{43}$$

Regarding the precondition $r_i$, the value of each pressure sensor is numerically superimposed, which is different from the previous admittance control. Admittance control requires the resultant force of its vectors. The precondition here is the situation whether the force is abnormal, the force is the sum of numerical superposition, $F_{left}$ and $F_{right}$ are as formula (44) and (45):

$$F_{left} = \sum_{i=1}^{8} |f_{li}| \tag{44}$$

$$F_{right} = \sum_{i=1}^{8} |f_{ri}| \tag{45}$$

$F_{left}$ and $F_{right}$ are classified and their thresholds are defined to maximize the sum of sensitivity and specificity. If the resultant force exceeds the critical point $F_{th}$, then it is judged that the fall defined by the precondition has occurred here, $r_i = 1$, otherwise $r_i = 0$, as formula(46):

$$\begin{cases} 1; & if\ F \geq F_{th} \\ 0; & if\ F < F_{th} \end{cases} \tag{46}$$

The flowchart of the improved Sequential Probability Ratio Test, namely Precondition and Limit threshold SPRT (PLT-SPRT) proposed in this work is shown in Fig. 13.

When the decision function $\Delta(k)$ is greater than or equal to the lower threshold $lnB$, it is no need to be corrected; when it is less than $lnB$, the decision function $\Delta(k)$ is corrected to $\Delta'(k)$ and the negative values are not accumulated. The optimized decision function has some improvement in eliminating the detection delay and reducing the false positive rate.

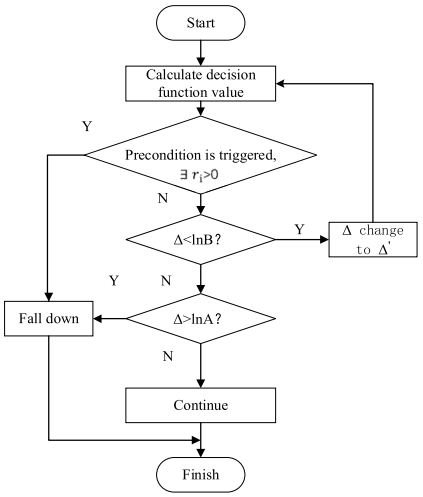



**FIGURE 13.** Structure of decision function modification.

## IV. SIMULATION AND EXPERIMENTS

Simulation experiments and real-world experiments with smart walkers are presented in this Section IV. The simulation experiment is composed of (1) experimental details and (2) simulation results. The real-world experiments with the smart walker are composed of (1) experimental details and (2) experimental results. Section C is a discussion of the above experimental process and results.

### A. SIMULATION EXPERIMENTS

#### 1) EXPERIMENTAL DETAILS

To simplify the model, the simulation experiments in this study are all on the premise of walking forward, without considering the lateral movement. To obtain the real admittance control model of the smart walker, the system identification method of the two-parameter model is used for parameter identification through Matlab software. The sampling signal of the handle sensors is shown in Fig. 14:

The real force sensor handle is shown in Fig. 14(a), it is fixed on the smart walker and connected to the sampling circuit board through the cable. To accurately and quickly detect the stable voltage signal of the force sensor and reduce noise interference, we use a third-order Butterworth low-pass filter with a 10kHz passband, as shown in Figure 14(b). After the voltage signal is filtered, it is amplified by an operational amplifier. The amplified voltage signal is calibrated, and finally, the force value is obtained.

In Fig. 14(c), the brighter area represents the greater the force on the area, and vice versa; the higher graph in Fig. 14(d) represents the greater the force on the area, and vice versa. It can be shown from the color, the red area is the most stressed. The intended velocity of the upper limb and the movement displacement of the lower limb are obtained by analyzing the data collected by the sensors.

The transfer function of the walker is formula (47):

$$G(s) = \frac{0.01}{(0.2s + 3)(0.5s + 2)} = \frac{V(s)}{F(s)} \tag{47}$$

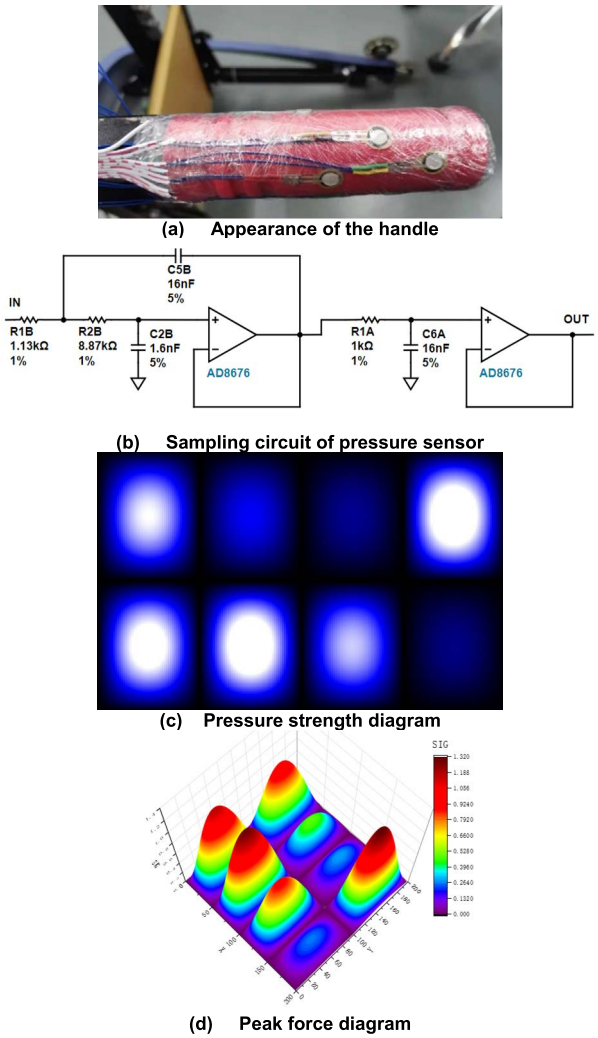



**FIGURE 14.** Schematic diagram of sampling signal of the upper limb force sensor.

Input the unit step signal, and get the response signal by the inverse Laplace transformation as formula (48):

$$y(t) = 0.0167 - 0.0277 * e^{-4t} + 0.0061 * e^{-15t} \quad (48)$$

Plot this response curve, according to the two-parameter model as formula (12):

$$\begin{cases} k = y(\infty) \\ A_0 = \int_0^{\infty} [k - y(t)]dt = T \cdot k \end{cases} \quad (49)$$

Calculate the values of $T$ and $k$, there is formula(50):

$$\begin{cases} k = 0.0167 \\ T = 0.3167 \end{cases} \quad (50)$$

According to the previous, the values of $T$ and $k$ are already known, which means $M$ and $C$ values of the admittance control model are known, the admittance control model is identified, and the identified response curve is drawn:

After obtaining the admittance control model, the real use of the walker is simulated. The force applied by the user while

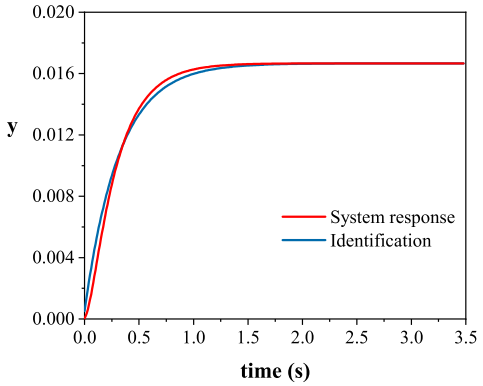



**FIGURE 15.** System response curve and identification curve.

using the walker is not a standard step signal, so Gaussian noise is added to the force signal, and the force gradually increases from 0N to 20N. The speed response curve of the walker is consistent with the above theoretical analysis curve as shown in Fig. 16.

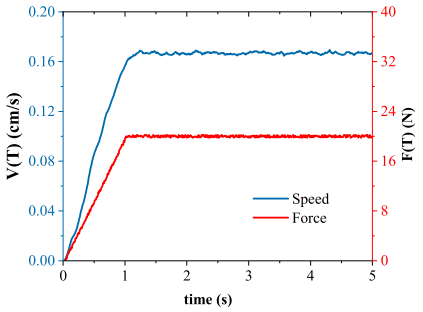



**FIGURE 16.** Admittance control test chart.

The lower limb uses ranging sensors matrixs to detect the swing of the user's legs. According to the previous analysis, it can be seen that the user's legs swing alternately when moving forward. Combined with the characteristic that the data collected by the sensor is close to the sinusoidal curve, the simulated sensor data is shown in the following Fig. 17:

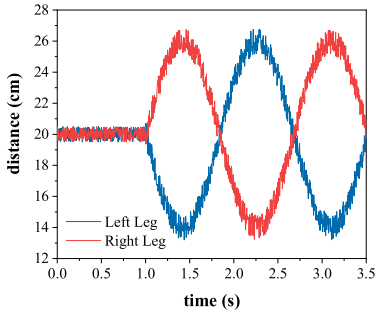



**FIGURE 17.** Ranging sensor data collection curve.

The displacement of the $Y$-axis gradually increases as the user walks forward. The displacement calculated by the sensors and the upper limb velocity calculated by the admittance control are used as the observed value, and the predicted value

is calculated according to the Kalman filter algorithm, the multi-sensor data fusion is carried out, and then the optimal estimated value is obtained after data fusion. Compared with the traditional Kalman filter, the $Y$-axis displacement results are as Fig. 18.

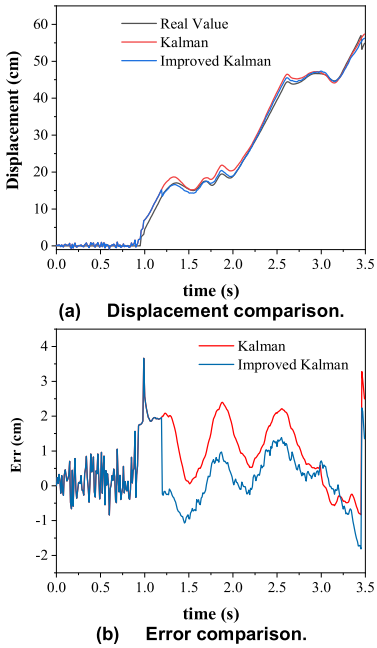



**FIGURE 18.** Comparison of kalman filter effect.

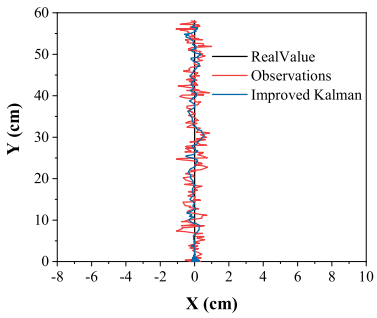



**FIGURE 19.** Displacement after multi-sensor data fusion.

The overall displacement of the user is as follows:

When the user falls forward, the grip strength of the upper limbs increases, and the distance between the legs of the lower limbs and the smart walker increases, as shown in Fig. 20(a) and Fig. 20(b).

### 2) SIMULATION RESULTS

We sampled a set of data using the sensors of the smart walker. This set of data is composed of thousands of time-series signals with a sampling interval of 10ms, that is, 100 data per second. Based on the actual fall time, 300 data are taken forward, 50 data are taken backward, and a total of 350 data are intercepted for analysis.

The sampling data for the first 3 seconds is the same as that of normal walking. After 3rd second, fall occurs, the

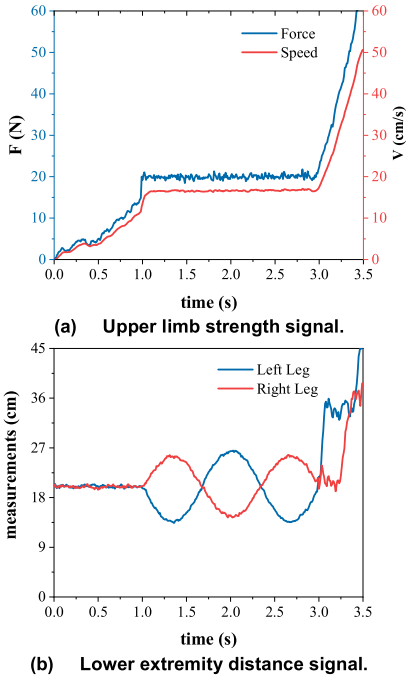



**FIGURE 20.** Upper and lower limb data for forwarding falls.

interaction force of the upper limbs increases, the intended speed increases, and the lower limb sensors detect that the distance between the walker and the legs increases, which means fall happened. At this time, according to PLT-SPRT, the null hypothesis and the alternative hypothesis are set in advance as the expected speed when walking normally and the expected speed when falling after many experiments, calculate the likelihood ratio and optimize the decision function. The detection results are divided into two types, the first type is the fall that does not trigger the precondition, and the second type is the fall that triggers the precondition.

A series of tests that the user walks with the smart walker shows that the walker can quickly detect the user's fall state and turn on the alarm as shown in Fig. 21.

As shown in Fig. 21, this study intercepted data for the 3 seconds before the fall and 0.5 seconds after the fall. In the first 3 seconds, the value of the decision function fluctuates slightly, but it is still always less than the threshold $lnA$, and no fall occurred. In the interval of 3 to 3.5 seconds, the decision function value is detected to be greater than the threshold, the fall is successfully detected.

Fig. 22 shows the user's normal advancement of the walker during the fall detection process. When one hand suddenly

**TABLE 1.** Value of the key variable of Figure 21.

| Symbol | Meaning | Numerical value |
|---|---|---|
| $lnA$ | Decision value cap | 2.9444 |
| $lnB$ | Decision value lower bound | -2.9444 |
| $\Delta t$ | Sampling interval | 10 ms |
| $N$ | Sample size | $3.5/\Delta t$ |

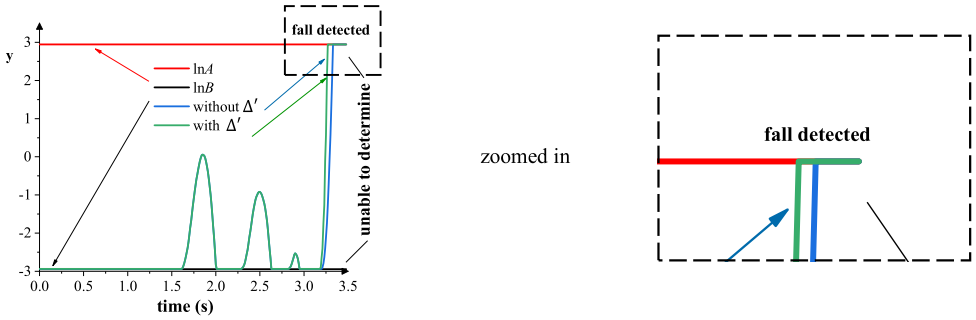



**FIGURE 21.** Sequential probability ratio test result without triggering precondition.

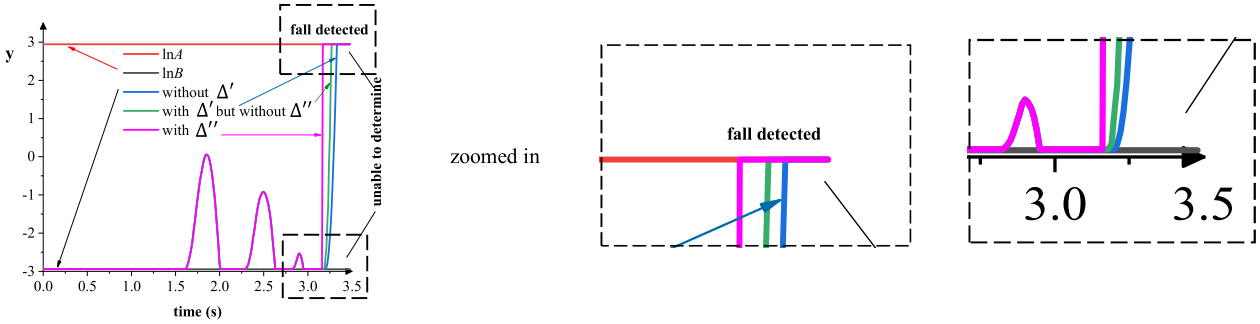



**FIGURE 22.** SPRT results with triggering precondition. Blue Curve is traditional SPRT, Green Curve is SPRT with optimized decision function, and Purple Curve is optimized SPRT with preconditions and weighting algorithm.

**TABLE 2.** Three results of figure 22.

| Test | Algorithm | Curve color | Time of fall detected | Difference from the shortest time Δt |
|---|---|---|---|---|
| $\Delta''$ | PLT-SPRT | Purple | 3.18 s | 0 ms |
| $\Delta'$ | SPRT with optimized decision function | Green | 3.28 s | +100 ms |
| $\Delta$ | Traditional SPRT | Blue | 3.34 s | +160 ms |

leaves the handle, the whole body falls. In this process, the preconditioning threshold is triggered first, and after that, the SPRT algorithm detects the fall. According to (40), $r = 1$, $\Delta'' = lnA$, when the preconditioning threshold is triggered, the algorithm has already determined the user's fall state.

According to the results in Fig. 21, Fig. 22, and Table 2, it can be seen that the optimized decision function algorithm detects falls faster than the original SPRT method, with lower latency, and the algorithm can provide a reliable guarantee for the safety of the smart walker. Some fall detection methods require a large amount of computation, such as the research of E. Auvinet *et al.* [17], and the different equipment used will have a greater impact on the speed, therefore there is no comparison of the detection speed with other methods in this paper.

### B. EXPERIMENTS USING SMART WALKER

#### 1) EXPERIMENTAL DETAILS

All actual data used in this study, including force and distance data, are obtained from 4 graduate students in the research group, and not from elderly patients in outpatient clinics.

The experiments are carried out in the laboratory open area of the institute, and the above 4 participants are asked to perform more than 1800 trials. Each test will generate a set of time-domain signals, the actual situation of whether the user has fallen, and the corresponding judgment of the smart walker.

We record the user's actual fall as "actual fall" and the non-fall as "actual ADL". The smart walker judges the fall as "judge fall", and the non-fall as "judge ADL".

The main circuit board chip of the smart walker uses the stm32F4 series chip. The structure of the embedded system is shown in Fig. 23.

#### 2) EXPERIMENTAL RESULTS

The data of test results, as shown in Table 3, can be divided into 4 cases in total: True Positive, False Positive, True Negative, and False Negative, which are abbreviated as TP, FP, TN, and FN respectively.

The evaluation metrics sensitivity(SE) and specificity(SP), are calculated according to formula (51).

$$\begin{cases} SE = \dfrac{TP}{TP + FN} \\ SP = \dfrac{TN}{TN + FP} \end{cases} \tag{51}$$

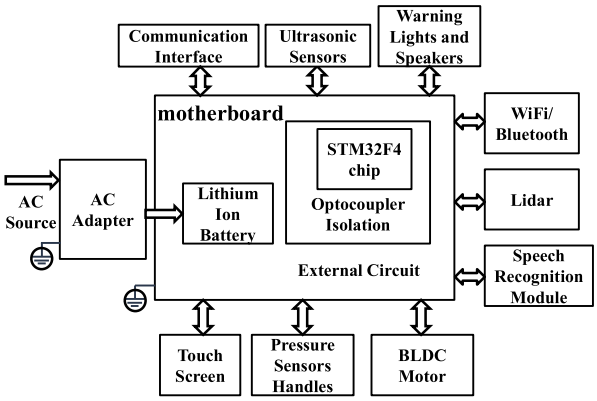



**FIGURE 23.** Structure of the embedded system.

**TABLE 3.** Confusion matrix of fall detection.

| | Judge fall | Judge ADL |
|---|---|---|
| Actual fall | TP | FN |
| Actual ADL | FP | TN |

Another evaluation metric accuracy (AC) is obtained according to formula (52).

$$AC = \frac{TP + TN}{TP + TN + FP + FN} \tag{52}$$

In addition, we also classify the numerical superposition ensemble F by analyzing the antecedent conditions and then determine the classification threshold. the ROC statistical analysis curve is shown in the following Fig. 25. The relevant data are presented in Table 4.

The ROC curves in Fig.25 show the classifier sensitivity versus 1-specicity at different detection thresholds, and different sensitivity and specicity combinations can be found.

**TABLE 4.** Sensitivity and specificity of fall detection precondition.

| Parameters | AUC | Std. Error | 95% CL | SE | SP | AC |
|---|---|---|---|---|---|---|
| Left | 0.9823 | 0.0028 | 0.976 to 0.987 | 0.981 | 0.937 | 0.961 |
| Right | 0.9901 | 0.0021 | 0.986 to 0.994 | 0.928 | 0.969 | 0.949 |

ESPRT[50] researchers and Chaos theory[51] researchers used DARPA 2000 as test data, and Luca *et al.* [14] used FARSEEING repository as test data, and other researchers used data collected by their own equipment. Most studies without hardware used standard datasets for testing, while studies with embedded devices or other types of hardware devices mostly use the data that generated by the research team. This research use data collected by sensors on the smart walker.

The classifier of Luca *et al.* [14] can improve its' sensitivity, at the expense of increasing false alarms to very high values (more than five per monitored hour), therefore, the SE data is a reasonable range.

To further validate the implementation of PLT-SPRT, we compare its results with those of other the-state-of-art methods as shown in Table 5. PLT-SPRT has an accuracy rate ranging from 94.9% to 96.1% when used with a smart walker. The average the accuracy is 95.5%, and our fall versus ADL detection performance is 92.8% to 98.1% sensitivity and 93.7% to 96.9% specificity, which is better than the others.

### C. DISCUSSION

In this paper, we propose a robust algorithm for detecting falls using low-cost sensors. Even if the handle is covered with many sensors, the cost is still very low. It is suitable for various indoor and outdoor environments, and will not cause additional detection obstacles due to insufficient light.



(a) Case of holding handles.



(b) Case of not holding handles.

**FIGURE 24.** Laboratory experiments of falling behavior.

**TABLE 5.** Comparison with other approaches.

| Method | Accuracy | SE | SP |
|---|---|---|---|
| **PLT-SPRT** | **94.9% ~ 96. 1%** | **0.928 ~ 0.981** | **0.937 ~ 0.969** |
| ESPRT[50] | 89.6% ~ 93.2% | 0.914 ~ 0.957 | 0.922 ~ 0.958 |
| Luca et al.[14] | NA | >0.8 | NA |
| Multiple Cameras[17] | NA | 0.947(6 Cam) | 0.956(6 Cam) |
| Chaos theory[51] | NA | 0.88 to 0.94 | 0.55 to 0.95 |
| Cheng et al. [52] | NA | 0.9533 | 0.9766 |
| Fang et al. [53] | NA | 0.7378 | 0.7722 |

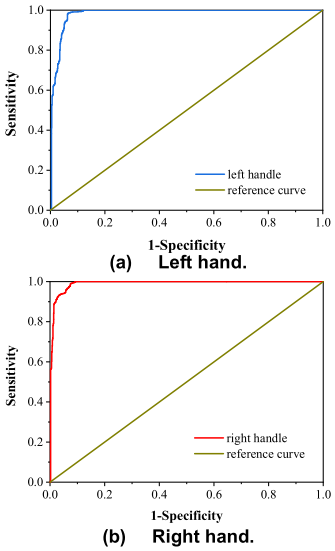



**FIGURE 25.** ROC curves of the left hand and right hand.

In addition, this algorithm can be easily generalized to detect falls in other directions, such as left and right, by a Multi-level SPRT approach, but it still should be validated.

All experiments simulating falls are done by healthy people using the smart walker. However, when a healthy person simulates a fall, this is different from an elderly or disabled person. We cannot compensate for this difference at present. It needs to be approved by an ethics committee and then select patients for clinical trials, which will cost a lot, and it is difficult to invite people with lower extremity defects, especially registered patients, to participate in a large number of trials. This is the limitation of our previous study. Two elder gentlemen are invited to do several independent simulated fall detections, and the smart walker can quickly and accurately detect the fall behavior. This shows that the data of young people is valid.

In addition, some factors can affect the accuracy of fall detection.

First, the strength and angle with which the user holds the handle of the smart walker can affect the accuracy of fall detection. When various users, especially their weight are very different, the impact on the handle has obvious differences. In addition, the height of the user will affect the

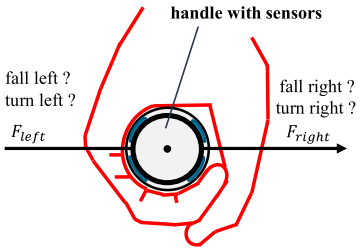



**FIGURE 26.** Research in other directions.

accuracy of detection. In the sensor signal map of the user's lower limbs as shown in Fig. 12(b), the user's waist, knees, and upper ankles are aligned with the corresponding sensors of the smart walker. If the user's height is too high, the signal obtained by the sensor at the waist of the walker may be the user's crotch, and the signal obtained by the sensor at the knee may be the user's lower leg. This may affect the accuracy of fall detection.

In addition to the above, if the user uses the smart walker for many years, the rebound characteristics of the thin-film pressure sensors on the handle may change, which will affect the sampling accuracy of the sensors. That will affects the accuracy of fall detection.

Moreover, when the user is not fully ready to use the smart walker, for example, the user just wakes up in the morning and falls while moving to the smart walker, the algorithm cannot detect the fall. This may be a big loophole before smart walkers become a good helper for users. To solve this problem, we will install a camera on the smart walker.

If the elderly are very healthy, and suddenly release the handle or directly leave the walker under normal circumstances, the algorithm will also report it as a fall. For this problem, we will add state discrimination later. For instance, in the case of underreporting, the algorithm will determine a state of ''lost-user'' instead of fall.

In the future, we will introduce cameras to improve fall detection accuracy and complete multimodal detection under ideal lighting conditions. Furthermore, we will also study fall prediction and steering prediction around the new handle of the smart walker. The users of smart walkers are elderly people with inconvenient legs or feet, who often do not have much strength when using mechanical tools. The user's turning intention and falling intention need to be distinguished by a suitable algorithm. Next step, we will study the judgment of steering intention and complete the steering of the smart walker rely on motor. Therefore, we will specialize in side steering prediction and fall detection.

## V. CONCLUSION

Among the traditional fall detection method, image recognition methods cannot be applied to all lighting environments. On the other hand, the single-line LiDAR is not suitable for outdoor scenes, and multi-line LiDAR is expensive. Moreover, the above algorithms are all complex, and many algorithms cannot be implemented on the smart walker. This

research proposes a novel fall detection system, and a fall detection algorithm based on the PLT-SPRT method by using multi-sensor data fusion of the upper and lower limbs of the smart walker. To complete this system, a new kind of handle is built for collecting upper limb signals.

To detect falls, a system identification algorithm based on a two-parameter model is firstly proposed to identify the admittance control model of the walker system and analyze the swing state of the user's lower limbs to establish a kinematic model. Furthermore, a fall detection algorithm based on the PLT-SPRT method is proposed, and the decision function is optimized to improve the delay of the traditional SPRT method. Experimental results show that the proposed algorithm can detect falls quickly and effectively, providing an effective and reliable solution for the fall detection of smart walkers.

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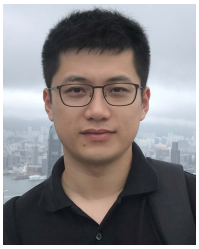
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